

GEOGRAPHY 03 — VOLCANISM AND EARTHQUAKES / INDIA SEISMIC ZONES

ENDOGENIC SYSTEM: PLATE MOTION → EARTHQUAKE SOURCE GEOMETRY, DEPTH → SEISMIC WAVES, SHADOW ZONES → GLOBAL SEISMIC BELTS, CASCADING → VOLCANO ANATOMY, MAGMA CONTROLS → VOLCANIC SETTINGS, GLOBAL BELTS

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

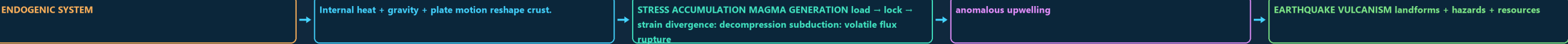
Approval: FALSE • Pending user review • source generation g7 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 ENDOGENIC SYSTEM: PLATE MOTION, STRAIN RELEASE AND MAGMA GENERATION

ENDOGENIC SYSTEM PLATE MOTION, STRAIN RELEASE AND MAGMA GENERATION INTERNAL HEAT + GRAVITY + PLATE MOTION RESHAPE CRUST. STRESS ACCUMULATION MAGMA GENERATION LOAD STRAIN DIVERGENCE DECOMPRESSION SUBDUCTION VOLATILE FLUX RUPTURE SLIP PLUME



DEFINITION / COMPONENTS

- ENDOGENIC SYSTEM
- Internal heat + gravity + plate motion reshape crust.
- STRESS ACCUMULATION MAGMA GENERATION load → lock → strain divergence: decompression subduction: volatile flux rupture → slip plume

TRIGGER / MECHANISM

- anomalous upwelling
- EARTHQUAKE VULCANISM landforms + hazards + resources
- FAULT PAIRS tension → normal

EFFECT / INTERVENTION

- compression → reverse/thrust
- shear → strike-slip
- Plate motion loads a region; sudden fault rupture is the immediate quake source.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Plate tectonics gives the spatial grammar of seismicity and volcanism, but the examiner still expects the immediate mechanics—fault slip for an earthquake and magma ascent for an eruption.

01

STAGE 01 EARTHQUAKE SOURCE GEOMETRY, DEPTH AND SEISMOGRAM LOGIC

EARTHQUAKE SOURCE GEOMETRY, DEPTH AND SEISMOGRAM LOGIC SURFACE STATION A ----- EPICENTRE ----- STATION B FOCUS HYPOCENTRE RUPTURE BEGINS ON FAULT PLANE SEISMOGRAM ARRIVAL ORDER P FIRST S SECOND LOVE AND RAYLEIGH LATER

P-S INTERVAL AT ONE STATION GIVES SOURCE DISTANCE THREE



SYSTEM / DEFINITION

- SURFACE station A ----- epicentre ----- station B focus / hypocentre rupture begins on fault plane
- SEISMOGRAM ARRIVAL ORDER
- P first → S second → Love and Rayleigh later

PROCESS / MECHANISM

- P-S interval at one station gives source distance three or more distance circles estimate the epicentre
- DEPTH CLASS shallow: 0-70 km
- intermediate: 70-300 km

SPATIAL / INDIA PATTERN

- deep: 300-700 km
- SEQUENCE FIREWALL foreshock, mainshock and aftershock are relational labels
- a swarm has no single clearly dominant event.

HAZARD / LIMIT / RESPONSE

- Epicentre is a vertical projection, not necessarily the worst-damaged place.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEQUENCE FIREWALL foreshock, mainshock and aftershock are relational labels.

02

STAGE 02 SEISMIC WAVES, SHADOW ZONES, MAGNITUDE AND INTENSITY

SEISMIC WAVES, SHADOW ZONES, MAGNITUDE AND INTENSITY MAIN CLUE SOLID, LIQUID CORE REFRACTION TRANSVERSE SHEAR SOLID ONLY LIQUID OUTER CORE HORIZONTAL SHEAR

WAVE	MOTION	MEDIUM	MAIN CLUE
P	compression	solid, liquid	core refraction
S	transverse shear	solid only	liquid outer core
Love	horizontal shear	surface	lateral shaking
Rayleigh	rolling ellipse	surface	mixed motion
P shadow about 103-142 deg	direct S shadow beyond about 103 deg		

DEFINITION / COMPONENTS

- MAIN CLUE
- solid, liquid
- core refraction
- transverse shear
- solid only

TRIGGER / MECHANISM

- liquid outer core
- horizontal shear
- lateral shaking
- rolling ellipse
- mixed motion

EFFECT / INTERVENTION

- P shadow about 103-142 deg
- direct S shadow beyond about 103 deg
- MAGNITUDE: one source measure; Mw uses seismic moment moment = rigidity x fault area x average slip
- 1 local magnitude gives about 10x amplitude and 31.6x energy
- INTENSITY: place-specific observed effects; many values for one event source x path x site x exposure x

- a swarm has no single clearly dominant event

STAGE 02 SEISMIC WAVES, SHADOW ZONES, MAGNITUDE AND INTENSITY

HORIZONTAL SHEAR

- P shadow about 103-142 deg
- direct S shadow beyond about 103 deg
- MAGNITUDE: one source measure; Mw uses seismic moment moment = rigidity x fault area x average slip
- 1 local magnitude gives about 10x amplitude and 31.6x energy
- INTENSITY: place-specific observed effects; many values for one event source x path x site x exposure x vulnerability / capacity → damage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Focal depth is a source descriptor, whereas disaster severity is a relational outcome produced by source, path, site, exposure and vulnerability.

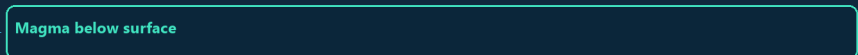
STAGE 03 GLOBAL SEISMIC BELTS, CASCADING HAZARDS AND TSUNAMI SEQUENCE

BASALTIC RIDGES

complex convergence

STAGE 04 VOLCANO ANATOMY, MAGMA CONTROLS, LANDFORMS AND ERUPTION STYLES

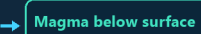
VISCOSITY UP



- First wave need not be largest; withdrawal need not occur before every event.
- Risk varies with bathymetry, bay shape, shelf, exposure and evacuation.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Depth distribution is the diagnostic map layer: only a descending slab generates a systematic inclined belt from shallow trench earthquakes to intermediate and deep events.

VISCOSITY UP



EXAM TRAP

- cinder cone
- fissure plateau
- STATUS: active
- extinct; status is not a compulsory life cycle.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Active status concerns eruptive capability; shield/composite concerns form; Hawaiian/Plinian concerns behaviour—UPSC can mix these axes in close options.

RING OF FIRE

[illegible]

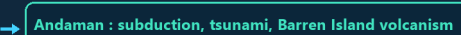
EXAM TRAP

- BENEFITS: new crust/land
- soil parent
- geothermal energy
- Impact converges through wind, relief, drainage, exposure and preparedness.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.

INHERITED RIFT AND FAULT REACTIVATION

ALLUVIUM, AMPLIFICATION AND HIGH EXPOSURE



HAZARD / LIMIT / RESPONSE

- non-structural safety → quality control → retrofit → drills
- Early warning detects rupture after it starts; it is not prediction.

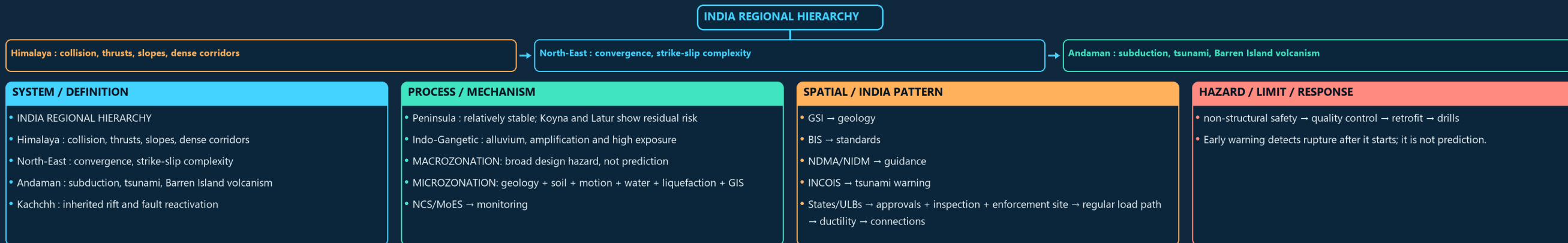
- geothermal energy
- Impact converges through wind, relief, drainage, exposure and preparedness.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.

STAGE 06 INDIA SEISMIC GEOGRAPHY, ZONATION, INSTITUTIONS AND RESILIENCE STACK

INHERITED RIFT AND FAULT REACTIVATION

ALLUVIUM, AMPLIFICATION AND HIGH EXPOSURE



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Macrozonation tells a city the regional design problem; microzonation tells neighbourhoods how local ground modifies that problem.

STAGE 07 HAZARD-RISK-RESPONSE AND MAINS ANSWER SPINE FOR SEISMIC INDIA

BASIN SEDIMENT + GROUNDWATER + SLOPE + LIQUEFACTION POTENTIAL

RISK VERDICT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: zonation and microzonation → construction and governance → conclusion.

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

ADVANCED 3 — CODE HIERARCHY, PERFORMANCE AND RETROFIT TRIAGE

FREQUENCY CONTENT MATTERS